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ci8718tl

Single 10-bit, 27MHz 3.3V/1.2V Video DAC

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Table of Contents

1.FEATURES..**3**

2.GENERAL DESCRIPTION..**3**

3.FUNCTIONAL DIAGRAM...**3**

4.SPECIFICATIONS...**4**

5.PIN DESCRIPTION..**6**

6.OPERATING MODES...**6**

7.TIMING DIAGRAMS...**8**

8.APPLICATION NOTES..**9**

Power Supplies...9
Digital Activity & Clock issues..9
Shielding...9
Analog Signals..9
ESD / Latch-Up..9

9.CELL ROUTING CONSTRAINS..**10**

10.CONNECTIONS TO THE IO PAD RING...**11**

11.PCB GUIDELINES...**12**

Power supply decoupling...12
Analog outputs configuration..12

1. Features

- **0.13um G 1P5M CMOS technology**
- **10-bit Resolution**
- **27 MHz Update Rate**
- **Full-scale Output Currents: 16/34mA**
- **Current Consumption:**
  - **26mA @ Ifs=17mA**
  - **45mA @ Ifs=34mA**
- **2.43 V to 3.6 V Analog Power Supply**
- **1.08 V to 1.32V Digital Power Supply**
- **Core Cell Area 0.25mm2**

APPLICATIONS:

- Any systems with an analog Video output

2. General Description

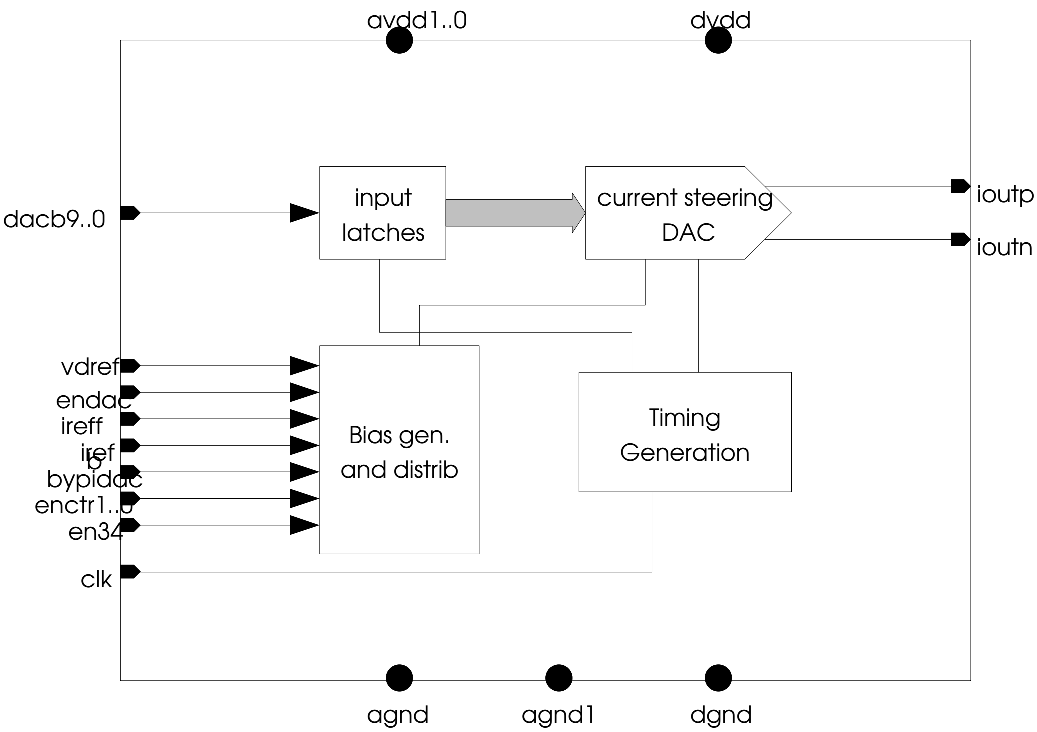
This 10-bit single Video DAC is designed for 0.13um 1P5M CMOS technology supplied at 3.3 V, Generic Process.

The DAC uses a segmented current steering architecture, combined with an improved two-dimensional centroid switching scheme, to achieve simultaneously high update rates, at least 10-bit of intrinsic static accuracy, and very good dynamic characteristics.

The circuit is a current output DAC designed to be loaded by double terminated 75 Ohm Lines.

An internal bandgap voltage reference, together with an external resistor, is used to set the full-scale current of the DAC. Alternatively, This full scale current can be set using an external current source.

3. Functional Diagram



4. Specifications

avdd = 3.3 V, dvdd= 1.2 V, Temp= 25°C, Fs=27MHz, 34mA full-scale current on a doubly terminated 75Ω load scheme, unless otherwise specified.

Parameter	Conditions	MIN	TYP	MAX	Unit
Target Process		0.13um G 1P5M CMOS + 3.3V IO devices			
Cell Area			0.25		mm^2
Junction Temperature		-40	50	125	ºC
Analog Biasing					
Reference Resistor (E96 Series)			1150		Ohm
Reference Current	For Ifs=34mA For Ifs=17mA		Ifs/32 Ifs/16		mA
Digital Inputs					
Logic Family	CMOS				
Input Logic Coding	Straight Binary: Bottom Scale: 000H Top Scale: 3FFH				
Analog Outputs					
Number of Channels			3		
Resolution			10		bit
Output Full Scale Current			17	34	mA
Output Compliance Range	Each single ended output			1.28	V
Resistive Load	Can be changed as long as output compliance range voltage is not exceeded	37.5	75		Ohm
Accuracy					
Offset Error			±1		%FS
Absolute Gain Error			± 5		%FS
DNL	Ifs=17mA Ifs=34mA			±0.5 ±1.0	LSB
INL	Ifs=17mA Ifs=34mA			±2 ±5	LSB
Timing Characteristics					
Update Rate		1		27	MHz
Clock duty cycle		45		55	%
Setup Time – T_{ST}		0.2	0.3	0.4	ns
Hold Time – T_H		0.7	0.8	1	ns
Propagation Delay – T_{PD}		0.4	0.5	0.8	ns
Startup Time	From Complete Shut-Down to normal operation		3		us
Power Supply					
Analog Supply Voltage – avdd		2.43	3.3	3.6	V
Digital Voltage Supply – dvdd		1.08	1.2	1.32	V
Analog Current Consumption	Ifs=17mA Ifs=34mA		27 46		mA mA
Digital Current Consumption	Fs=27MHz		4		mA

avdd = 3.3 V, dvdd= 1.2 V, Temp= 25°C, Fs=27MHz, 34mA full-scale current on a doubly terminated 75Ω load scheme, unless otherwise specified.

Parameter	Conditions	MIN	TYP	MAX	Unit
Dynamic Performance					
SFDR	Within a band of 400KHz to 5MHz, Fout=2.2MHz, Ifs=17mA, R_L=75Ω, Fs=27MHz Fout=2.2 MHz, Ifs=34mA, R_L=37.5Ω, Fs=27MHz		 60 47		dBc
SINAD	Within a band of 400KHz to 5MHz Fout=2.2MHz, Ifs=17mA, R_L=75Ω, Fs=27MHz Fout=2.2 MHz, Ifs=34mA, R_L=37.5Ω, Fs=27MHz		 57 46		dBc

5. Pin Description

Pin Name	I/O	Type	Function
iref	I/O	Analog	Reference Current. Output Current when using External Reference Resistor or Input Reference Current when using external current source
ireffb	I/O	Analog	Reference Current sensing node. Must be tied to iref pin as near as possible to the pad.
vdref	I	Analog	Reference Voltage supply
bypidac	I	Digital	External Reference Current Enable (active *HIGH*)
endac	I	Digital	Enable control pin to power up the DAC
en34	I	Digital	34mA output full-scale enabled (active *HIGH*)
enctr1..0	I	Digital	Enable control pins for analog biasing Test
dacb9...dacb0	I	Digital	Digital Input bits of the DAC
clk	I	Digital	Clock signal
ioutp, ioutn	O	Analog	Positive and Negative Output Currents for the DAC
avdd1..0, agnd	I/O	Analog	Analog power supply and ground
agnd, agnd1	I/O	Analog	Analog ground reference
dvdd, dgnd	I/O	Digital	Digital power supply and ground

6. Operating Modes

In the 'Internal Bias Mode' (**bypidac** set to LOW), the full-scale current is determined by an external resistor, connected from pin **iref** to **agnd**, biased with a replica of the internal bandgap reference voltage – *Vbg*. The current trough the external resistor is internally mirrored by a well-determined factor to define the full-scale current.

Alternatively, in the 'External Bias Mode' (**bypidac** set to HIGH), an external reference current of one fraction to the full-scale current must be applied from **iref** to the ground. The output current on pin **iref** is always positive (flowing out of the pin).

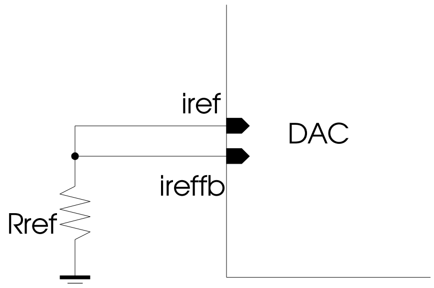
Mode	Description	Configuration		
		endac	en34	bypidac
0	Complete Power Down	0	X	X
1	Normal Mode, 17mA output full scale mode	1	0	0
2	Normal Mode, 34mA output full scale mode	1	1	0
3	DAC is active	1	X	0
4	DAC active, using external reference current	1	X	1

Mode 0 - Power Down
In this mode all the circuitry is in complete power-down. The power dissipation is reduced to its minimum value.

Mode 1 – Normal Mode, 17mA output full scale mode.
All channels are active.
In this mode, an external reference resistor must be used to define the internal reference current as represented in figure bellow. The internal reference current I_{REF} is defined as:

$$I_{REF} = \frac{V_{bg}}{R_{REF}}$$

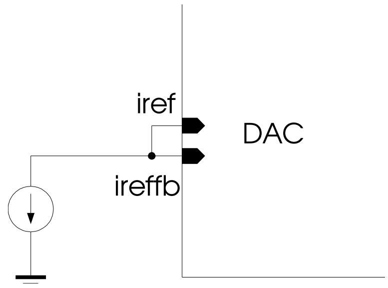
where R_{REF} is the value of the outside reference Resistor, which must be chosen according to specifications table and Vbg has nominally 1.21 Volt.



Mode 2 – Normal Mode
All channels are active, 34mA output full scale mode. The R_{REF} value must be chosen according to specifications table.

Mode 3 – DAC is active

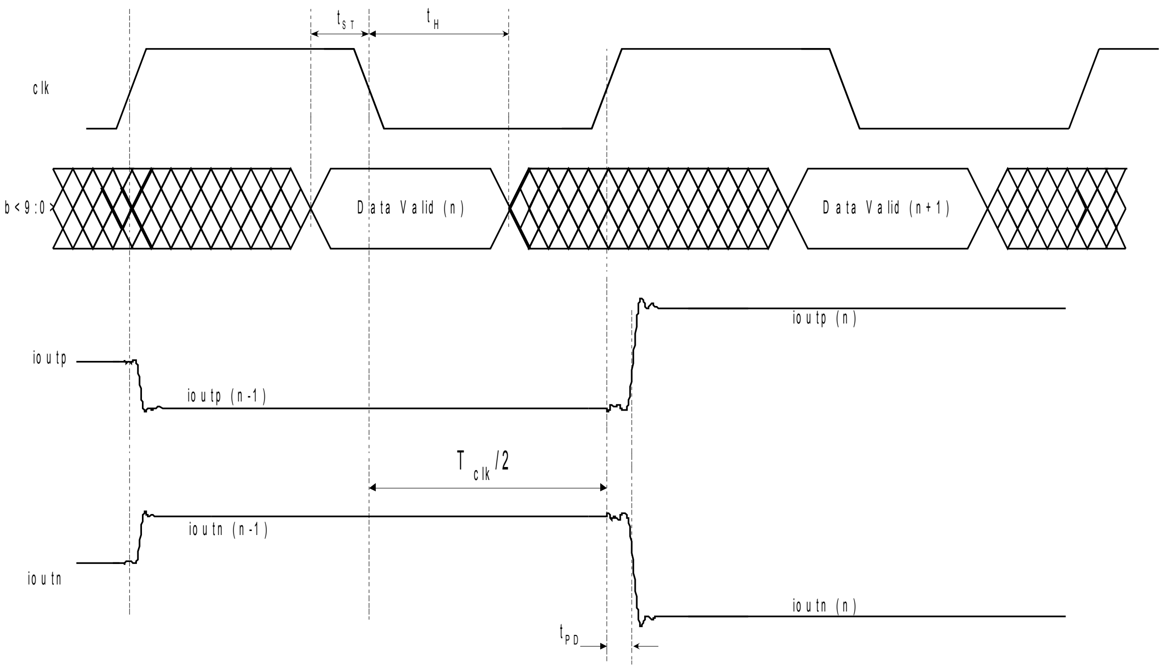
Mode 4 – External Reference Current
This mode can be used in conjunction with modes 1, 2 or 3.
In this mode, an external reference current must be applied in pin Iref flowing from pin to ground as represented in figure bellow.



7. Timing Diagrams

The DAC operates according to the following timing diagram.

The digital input word is latched on the falling edge of the clock signal. In the next rising edge of the clock, the latched data digital word is converted to its analog value at the outputs of the DAC.



Timing Diagram1 - Input and Output Timing Characteristics of the DAC

8. Application Notes

Power Supplies

- Use separate Pins and Pads to all core positive cell supplies. If this is not possible then:
  1. In case of Pin limited ASIC share power supply pins using doubling.
  2. In case of Pad limited ASIC share power supply Pads by merging the power supplies nets as close as possible to the Pad.
  3. Short all ground nodes (analog and digital) to a single ground path.
- If possible use double bonding to all Core supplies as recommended in the cell datasheet.
- Use separate rings between analog and digital core.
- Use minimum amount of power cuts between analog and digital ring according to ESD guidelines in order to decrease the capacitive coupling between the I/O rings. This is especially important when there are high frequency digital switching.
- In case of several analog power supplies, guarantee the same length from the analog core to the pads.

Digital Activity & Clock issues

- Ensure good quality low jitter clock according to each system specifications such that analog core performance is not compromised.

Shielding

- Use high resistive substrates for Mixed Signal ASIC.

Analog Signals

- Keep all analog signal routing as small as possible.
- Keep all differential signal routing as symmetric as possible
- Maintain the same wire length between analog core cell and Pad a in all differential type signals
- Shield all critical analog signals, especially the ones that are single ended type. Use analog ground as shielding to all analog signals

ESD / Latch-Up

All input / output elements connecting directly to pads, are designed according to the rules for ESD and Latch-Up protection.

Analog pads should not have a series resistance, in order not to deteriorate performance.

Minimize the number of power cuts between the analog and digital rings, in order to decrease the capacitive coupling between them. This is especially important when there is high-frequency digital switching. The ESD guidelines of the technology must, obviously, be respected

9. Cell Routing Constrains

The table below presents the signals whose routing must follow constrains.

Unless otherwise stated the analog signal shielding is made with each cell's agnd (or avdd as 2nd option).

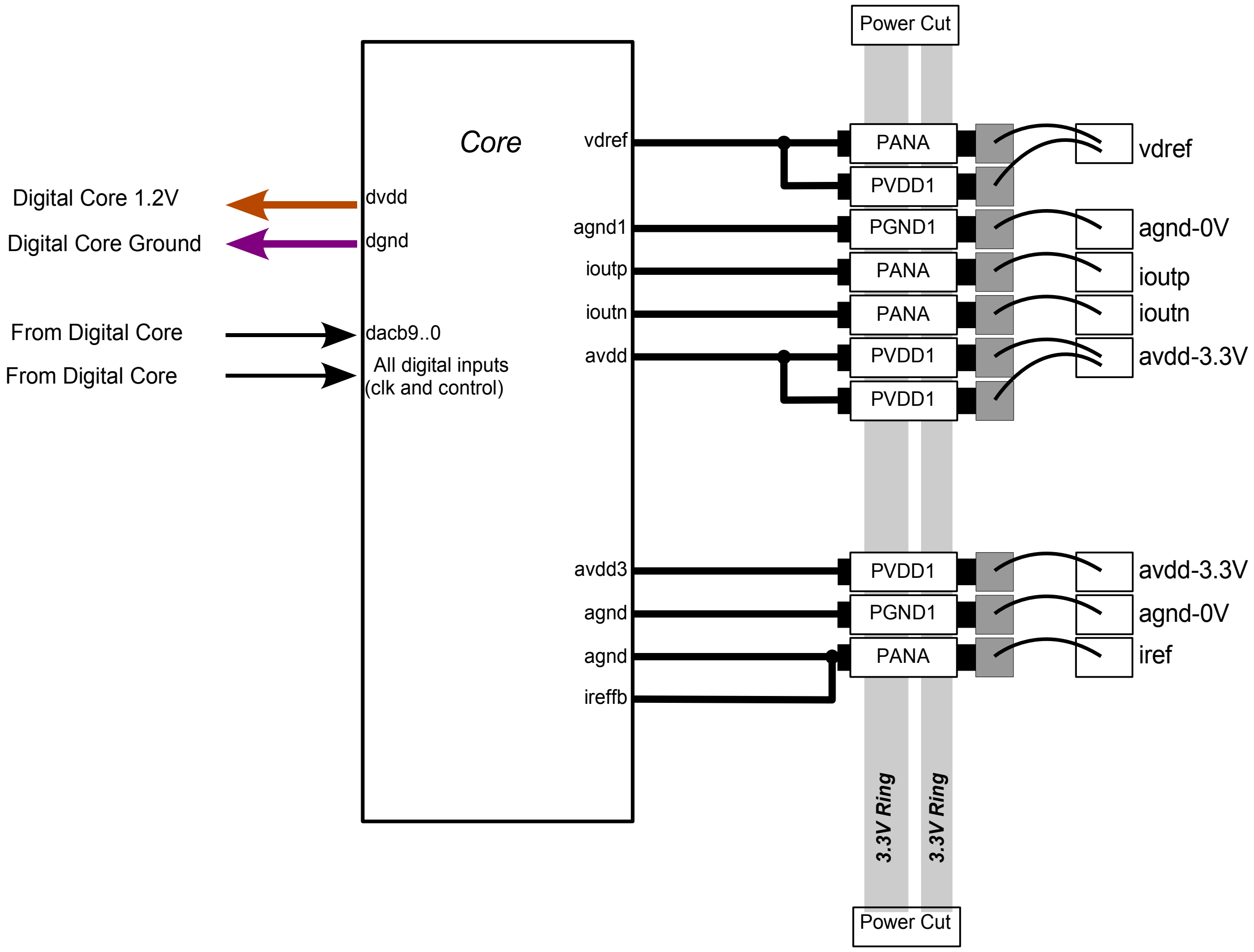
Cell Pin Name	Wire Resistance [ohm]	Wire Current [mA]	Recommendation
Power			
avdd3..0	<1	>34	n.a.
agnd	<5	>15	n.a.
dvdd	<10	>5	n.a.
dgnd	<10	>5	n.a.
agnd1	<3	>15	n.a.
References and Bias			
iref	<10	>2.5	Shield from any digital or analog signal.
ireffb	<10	>0.1	Shield from any digital or analog signal.
vdref	<3	>15	
output Signals			
ioutp(n)2..0	<1	>34	Shield from any digital signal or reference signal

n.a. – Not Applicable

10.Connections to the IO PAD ring

The figure below indicates how the connections between the cell and the IO PAD ring should be made. The figure presents the recommended situation, and the others are compromise solutions. The following notation is used:

- PVDD1 is a generic 3.3V Power PAD to bias both the IO ring and the core cell.
- PGND1 is a generic ground Power PAD to bias both the IO ring and the core cell.
- PANA is a generic Analog IO PAD without an internal series ESD resistor.

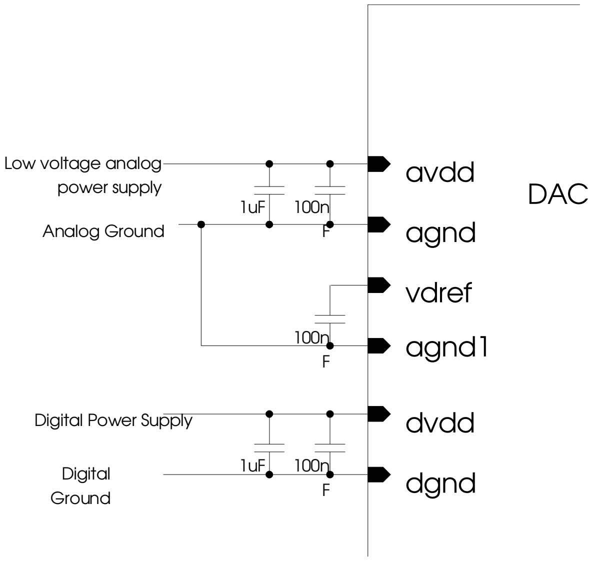


IO ring case – Separate Analog IO ring for this IP.

11.PCB Guidelines

Power supply decoupling

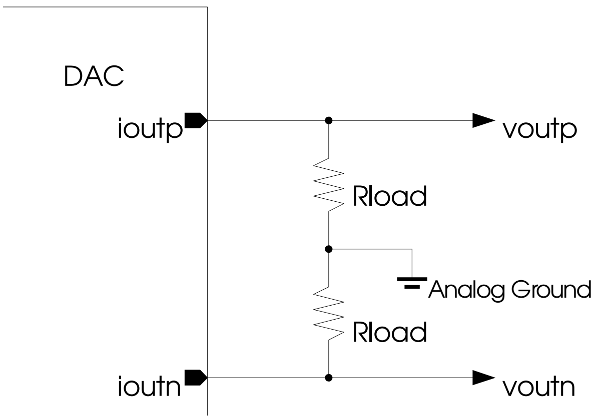
- Power supply decoupling should be done according to the following figure.



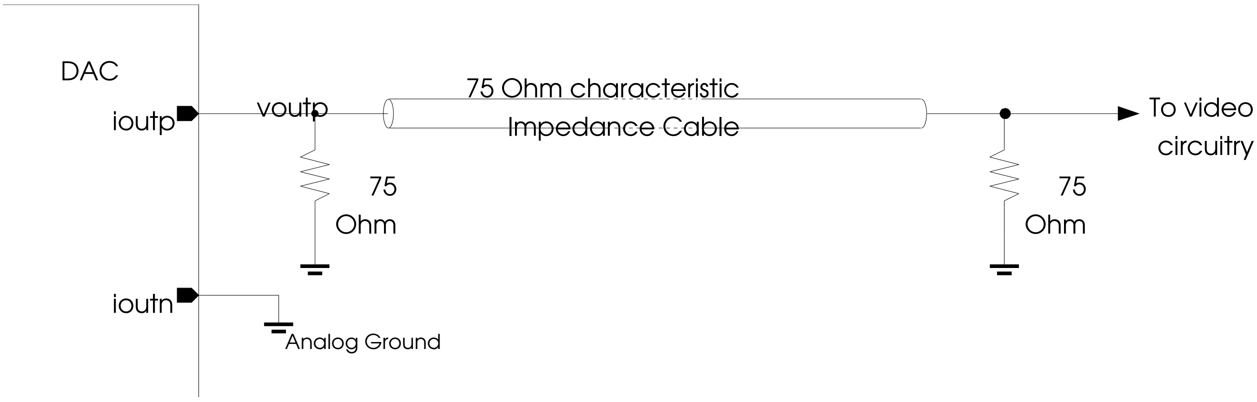
- It is recommended the 100nF capacitors to be placed as close as possible to the chip.
- At least the small 100nF should be good quality ceramic capacitors.

Analog outputs configuration

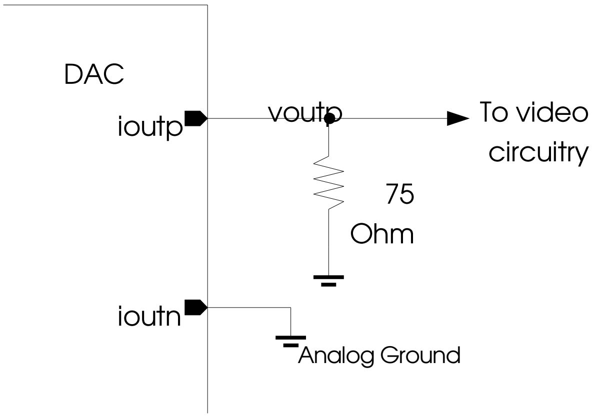
- Simple configuration to implement output current to voltage conversion.



- Most common configuration to convert current to voltage in a video application using a transmission line (double termination, en34=1):



- Alternative configuration to convert current to voltage in a video application without transmission line (single termination, en34=0):



Revision History

Doc. Revision	Date	Prepared	Description
0.1	2007-01-26	A. Sequeira	ci87xxtl-single-da10b200vdac_SMK_DS_0v1 • Initial Release based on: ci8714tl_da10b200vdac_SMK_DS_1v1.sxw
0.1	2007-01-30	A. Sequeira	ci8718tl-single-da10b200vdac_SMK_DS_0v1 • Contract signed, Part number assigned, same as ci87xxtl-single-da10b200vdac_SMK_DS_0v1
0.2	2007-02-27	A. R. Leal	ci8718tl-single-da10b200vdac_SMK_DS_0v2 • Update of area to 0.25 mm2. 0.2mm2 would be the DAC area without biasing blocks included.
1.0	2007-03-30	A. R. Leal	• Final Version • Deleted sepcification “Gain Error (DAC to DAC matching)” as it is meaningless in a single channel DAC.